Ved Dalwadi

In this journal entry, I will share my experiences during the ITAI 1371 midterm project. In this project, my group and I worked with the Telco Customer churn dataset. This is a split assignment between all 3 members of the group; I will explain my part in handling the Data cleaning and test split training of the dataset. The reason for this reflection is to help show my thinking process and what I learned as I cleaned my first dataset.

When I first started this project, I immediately saw an issue in the data. After I had set up all necessary libraries, I first printed out df.info(). As I was scrolling through the columns, the Total Charges column seemed incorrect, as it was labeled as an object instead of an integer, which could cause problems if we later used these numbers to train the AI. After seeing this, I decided to look into the Total Charges column first. I saw that there were 11 missing values for the column. This may not have been a big deal for a person looking at the database, but any AI would get thrown completely off by a string and stop working entirely, as it's considered a string in the eyes of the AI. I used pandas to transform the strings into numbers, and for the empty rows I used errors='coerce' to turn them into NAN instead of leaving them blank. I then decided to fill any NAN with 0, even though I was taught that the median is the safest option for guessing. I believe that these customers just haven't been billed, as I also saw low tenure in these customers as well, which most likely means they haven't paid yet.

Another issue I saw with the dataset in this project is the customer ID column. At first look, it may just look like it's just identifying customers; however, in a project, unnecessary numbers are exactly what machine learning builders look out for every day. Unnecessary numbers like these can completely throw a model off and are not useful to the AI.

Module 5 taught us about one-hot encoding, and as I was dropping this column, this somewhat reminded me of this idea of keeping data in categorical columns as 0's and 1's to make sure no category got an unfair advantage. If we don't, we may make the AI believe the column labeled 2 is more important than 1 when it's not, while customerID is not that type of problem, both don't let the AI see unnecessary numerical data.

Another part of the cleaning process I handled was creating the before and after histograms for Total Charges. I plotted both the raw and cleaned datasets side by side and noticed that, because only 11 out of 7000 rows are blank, the shape was very similar to the raw data. This is a similar realization I come to in all of my labs, where I start to understand that little things matter just as much as the big picture. I have stated that small steps may not seem to have a major impact; however, these small changes do matter, as in the end it will increase the AI accuracy if you assume correctly, and without these small parts the entire machine could not learn properly. This shows that datasets basically look the same; they still have importance, no matter how small the change is, as it still affects the results.

After finishing cleaning the dataset, my next task was to split the data into 70/30 train/test sets. I remembered this process from the module 3 journal; this is also the module where I learned that every part of machine learning is important, and everything is just as important as the last. While thinking about the previous module, a question came to mind if someone did make a mistake in the code, would it affect other members of the group, and would they be able to notice? This is concerning, as if something can slip through multiple people and you decide that your data is ready for the AI and it’s not, it could lead to hundreds of hours wasted by one mistake in the beginning.

This project showed me that even though a data engineer can have all the same problems as a different data engineer, the way they solve these problems can determine whether the AI is great or horrible. In my module 3, I asked the question of how to understand an AI and not just plug in numbers until it works. These last 2 modules and this midterm have helped me reach this goal by showing me the critical thinking necessary to make decisions on what data to keep and what to get rid of. This midterm and previous modules helped me understand how the first step of machine learning is done, and I hope to learn the next steps in the next modules.

Reflective Journal: ITAI 1371 Midterm Project Nigel Bontiff

For this midterm project, my group worked with the Telco Customer Churn dataset. The main goal was to clean and prepare the dataset so it could eventually be used to train machine learning models that predict whether a customer is likely to cancel their service. My part of the project focused on one-hot encoding the categorical data, scaling important numerical columns, comparing the data before and after preprocessing, and using class balancing with Logistic Regression.

One thing this project helped me understand better is that raw data usually cannot be placed directly into a machine learning model. The Telco dataset had a lot of useful information, but most of the columns contained text values such as contract type, payment method, internet service, and whether the customer had different services. These values make sense to a person, but the model needs numbers to work with them. My first responsibility was to make those categories cleaner and easier to encode.

Some columns contained values like “No internet service” or “No phone service.” Technically, those values were different from a regular “No,” but they were still telling us that the customer did not have that service. I changed those values into a simpler “No” category using Python. This made the categories more consistent and prevented the encoding process from creating extra columns that were not really necessary. It also showed me that data cleaning involves thinking about what the values actually mean instead of only changing them because the instructions say to.

After cleaning the categories, I used one-hot encoding to convert the remaining categorical columns into numeric columns containing 0s and 1s. This was one of the more important parts of my section because the dataset had many object or text columns. One-hot encoding allowed the model to use information like contract type and payment method without incorrectly treating the categories as if they had a numerical ranking. For example, a month-to-month contract is not mathematically “less than” a two-year contract, so assigning them random numbers could give the model the wrong idea.

I also applied `StandardScaler` to `tenure`, `MonthlyCharges`, and `TotalCharges`. These columns were all numerical, but their values were on different ranges. Tenure was measured in months, while total charges could reach several thousand dollars. Without scaling, a model like Logistic Regression could be influenced more by a column simply because its numbers were larger. Standardizing the columns placed them on a more comparable scale.

The before-and-after graphs were helpful because they showed what scaling actually does. The general shape of each distribution stayed similar, but the values were moved onto a standardized scale. Before this assignment, I understood scaling mostly as a formula or code command. Seeing the graphs made it clearer that scaling does not erase the original pattern in the data. It changes the range and center of the values so the model can work with them more fairly.

Nigel Bontiff

Another part of the assignment was dealing with class imbalance. The dataset had more customers who did not churn than customers who did churn. This can be a problem because a model may learn to mostly predict the larger class and still appear accurate. I used `class_weight='balanced'` in the Logistic Regression model so the smaller churn class would receive more attention during training. I liked this method because it did not manually add, remove, or change rows in the dataset. Instead, it changed how much importance the model gave to each class.

One challenge was making sure the steps happened in the correct order. The professor instructed us to split the dataset first, perform preprocessing on the training data, and leave the original testing data untouched. My section depended on the earlier cleaning and train-test split being completed before I could begin. I had to make sure I used a copy of the training set instead of accidentally changing the original test data. This helped me understand why data leakage and proper workflow order matter. A model can look more accurate than it really is if information from the test set is used during preprocessing or training.

I also had to think carefully about which columns should be encoded and which columns should be scaled. Not every numeric-looking column should automatically be scaled, and categorical columns need a different type of preparation. This project made the difference between data types feel more practical than it did when we only discussed them in class.

My main contribution to the group was preparing the training data so it became cleaner, numeric, scaled, and ready for machine learning. I simplified repeated categories, one-hot encoded the text columns, scaled the selected numerical features, created before-and-after visualizations, saved the processed training dataset, and added class balancing to the Logistic Regression model. I also wrote an explanation of how these steps connect to the customer churn problem.

Overall, this project helped me see how the concepts from the Exploratory Data Analysis and Data Preparation modules fit together. EDA helps us understand what is happening in a dataset, while preprocessing turns that understanding into actual changes that make the data usable for modeling. My biggest takeaway is that the quality of the model depends heavily on how the data is prepared. Even a good algorithm will struggle if the data is inconsistent, unbalanced, or still contains text values that it cannot understand.

Monepha Mitchell

ITAI 1371 Introduction to Machine Learning

Professor Viswanatha Rao (Vishwa)

Assignment: Lab 03 Exploratory Data Analysis

Reflection Journal

Working on this midterm project has helped me better understand how important data preparation and exploratory data analysis are in the machine learning process. Before building any machine learning model, it is important to understand the data, identify any problems, and prepare it correctly because the quality of the data has a direct impact on the model's performance. My responsibility for this project was to complete the exploratory data analysis and feature engineering on the training dataset. I explored the data by generating summary statistics and creating visualizations for contract type, tenure, payment method, and monthly charges.

These visualizations helped me better understand customer behavior and recognize patterns that may influence customer churn. I also examined the class distribution of the target variable and learned why class imbalance is an important issue to address before training a machine learning model. Although the class balancing step will be completed after the categorical variables are one-hot encoded, I now understand why this process is necessary to improve model performance. Another valuable part of my work was feature engineering, where I learned how creating new features, such as grouping customers by tenure or counting the number of services they subscribe to, can provide additional information that may improve prediction accuracy.

Working as a group also made the project more manageable because we communicated regularly, discussed the assignment requirements, shared ideas, and helped each other understand different parts of the project while still completing our individual responsibilities. Overall, this project has strengthened my understanding of exploratory data analysis, feature engineering, and the overall machine learning workflow, and I feel much more confident applying these skills to future machine learning projects.